

SCIENTIFIC OVERVIEW

Xanomeline-trospium chloride (XTC) A Novel Paradigm in Schizophrenia

Andre R. Alexander

TALK TRACK

Welcome. Today, we discuss XTC, a potential first-in-class muscarinic agonist that represents the first non-D2-centric breakthrough in schizophrenia in over 50 years

Context & Disclosures

Professional Intent

This presentation was developed independently as a professional medical writing sample to demonstrate data synthesis and visual design capabilities.

No Conflicts

The author has no financial affiliations, conflicts of interest, or sponsorships to disclose regarding the products discussed herein. Data is from peer-reviewed sources.

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This deck is a self-directed clinical analysis. All data presented here is derived from public, peer-reviewed literature, primarily the EMERGENT clinical trial program.

Unmet Need in Schizophrenia



Current D2-Centric Antipsychotics Fail to Address Cognitive and Negative Domains

Current SOC relies on D2 blockade. Effective for positive symptoms but fails to address cognitive/negative domains.^{1,2,3}



Side Effects

Metabolic syndrome, EPS, and sedation can cause significant disability.⁴



The Gap

There is a critical need for treatments targeting negative symptoms with improved tolerability profiles.

1. Huhn M, Nikolakopoulou A, Schneider-Thoma J, et al. Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of adults with multi-episode schizophrenia: a systematic review and network meta-analysis. *Lancet*. 2019;394(10202). doi:10.1016/s0140-6736(19)31135-3
2. Cerveri G, Gesi C, Mencacci C. Pharmacological treatment of negative symptoms in schizophrenia: update and proposal of a clinical algorithm. *Neuropsychiatr Dis Treat*. 2019;15:1525-1535. doi:10.2147/NDT.S201726
3. Feber L, Peter NL, Chiochia V, et al. Antipsychotic drugs and cognitive function: a systematic review and network meta-analysis. *JAMA Psychiatry*. 2025;82(1):47–56. doi:10.1001/jamapsychiatry.2024.2890
4. Chow RTS, Whiting D, Favril L, Ostinelli E, Cipriani A, Fazel S. An umbrella review of adverse effects associated with antipsychotic medications: the need for complementary study designs. *Neurosci Biobehav Rev*. 2023;155:105454. doi:10.1016/j.neubiorev.2023.105454

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Despite decades of progress, we still struggle with 'the revolving door' of schizophrenia. While positive symptoms are managed, the functional disability of negative symptoms remains a massive unmet burden.

MOA: Selective Muscarinic Agonism

- ✓ **Xanomeline:** M1 and M4 mAChR-preferring agonist. Acts as both an orthosteric and allosteric agonist at the M4 receptor.⁵
- ✓ **Trospium:** Competitive antagonist at muscarinic cholinergic receptors.⁶
- ✓ **Blood-Brain Barrier:** Trospium does not cross the BBB, preserving central therapeutic effect.⁷

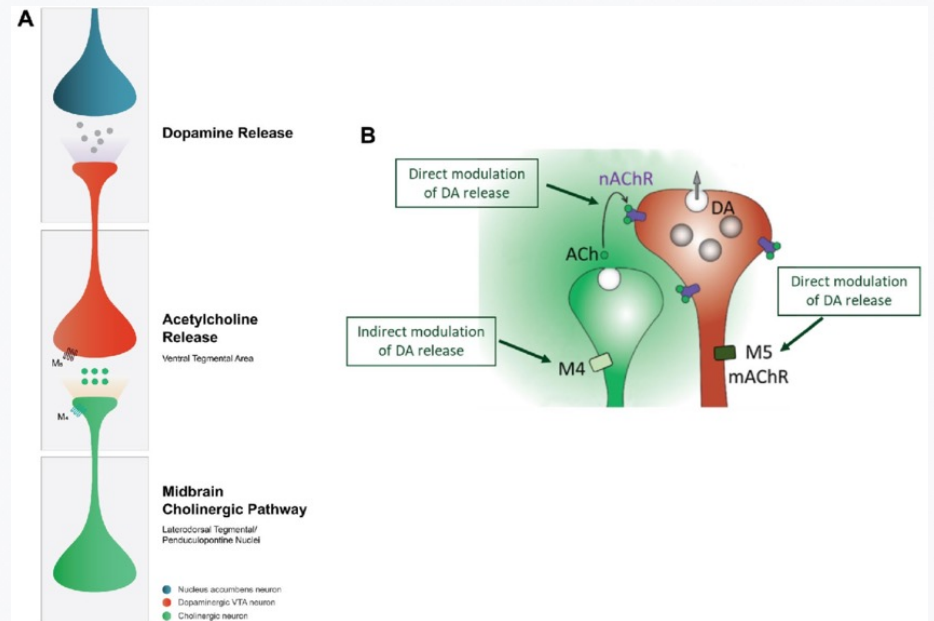


Fig. 3 **A** Midbrain muscarinic input increases dopamine release. **B** Modulation of dopamine release by acetylcholine via subtypes of nicotinic and muscarinic cholinergic receptors in the nucleus accumbens. Reproduced from Meyer JM, Correll CU. Increased metabolic potential, efficacy, and safety of emerging treatments in schizophrenia. *CNS Drugs*. 2023;37(7):545-570. doi:10.1007/s40263-023-01022-7 Licensed under CC BY-NC-ND 4.0. Unaltered image.

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The genius of XTC is the co-formulation. Xanomeline provides the central antipsychotic "punch" via M1/M4, while Trospium acts as a peripheral shield against GI side effects

5. Wessel, Pham V, Vuckovic Z, et al. Xanomeline displays concomitant orthosteric and allosteric binding modes at the M4 mAChR. *Nat Commun*. 2023;14(1). doi:10.1038/s41467-023-41199-5
 6. Guay DR. Trospium chloride: an update on a quaternary anticholinergic for treatment of urge urinary incontinence. *Ther Clin Risk Manag*. 2005;1(2):157-167. doi:10.2147/tcrm.1.2.157.62912

Study Design: EMERGENT-3⁹

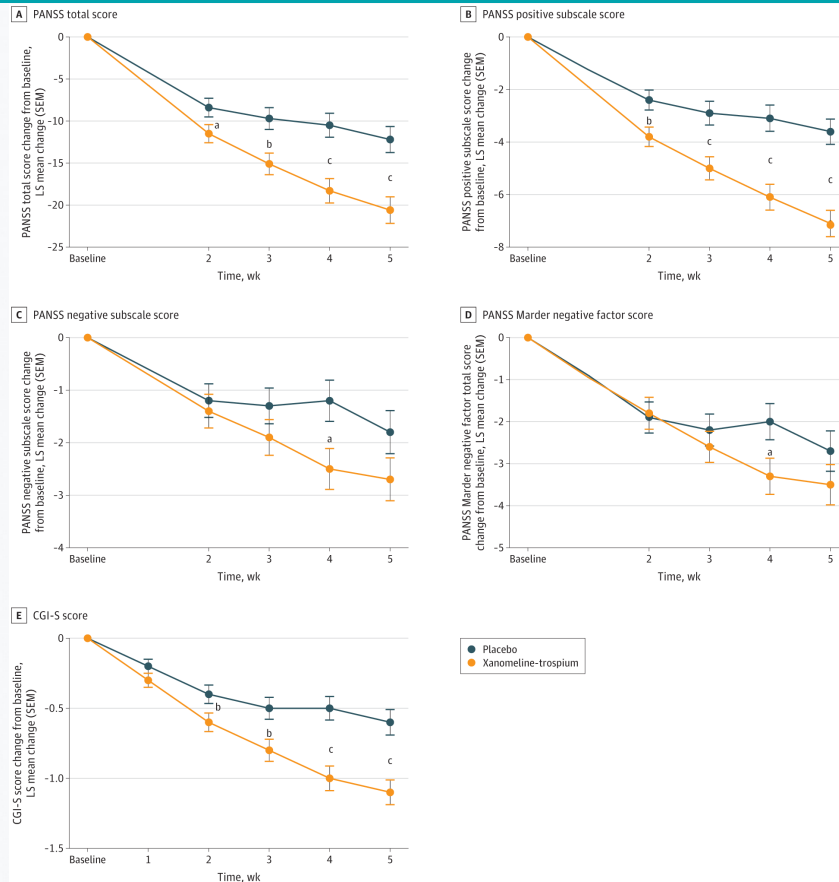
Feature	EMERGENT-3 Specifications
Study Type	Phase 3, Multicenter, Randomized, Double-Blind
Population	Adults (18-65) with Acute Schizophrenia Worsening
Primary Endpoint	Change in PANSS Total Score (Baseline to Week 5)
Sample Size	N = 256 Randomized (1:1 Ratio)
Dosing	Flexible titration up to 125mg Xanomeline / 30mg Trospium

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EMERGENT-3 was an acute 5-week inpatient trial. We specifically looked at patients in acute relapse to see if XTC could manage severe positive symptoms without direct D2 intervention

9. Kaul I, Sawchak S, Walling DP, et al. Efficacy and safety of xanomeline-trospium chloride in schizophrenia: a randomized clinical trial. *JAMA Psychiatry*. 2024;81(8):749-756. doi:10.1001/jamapsychiatry.2024.0785

Primary Efficacy: PANSS Total Score⁹



Significant Reduction

-8.4 Point Difference vs Placebo ($p < 0.001$)

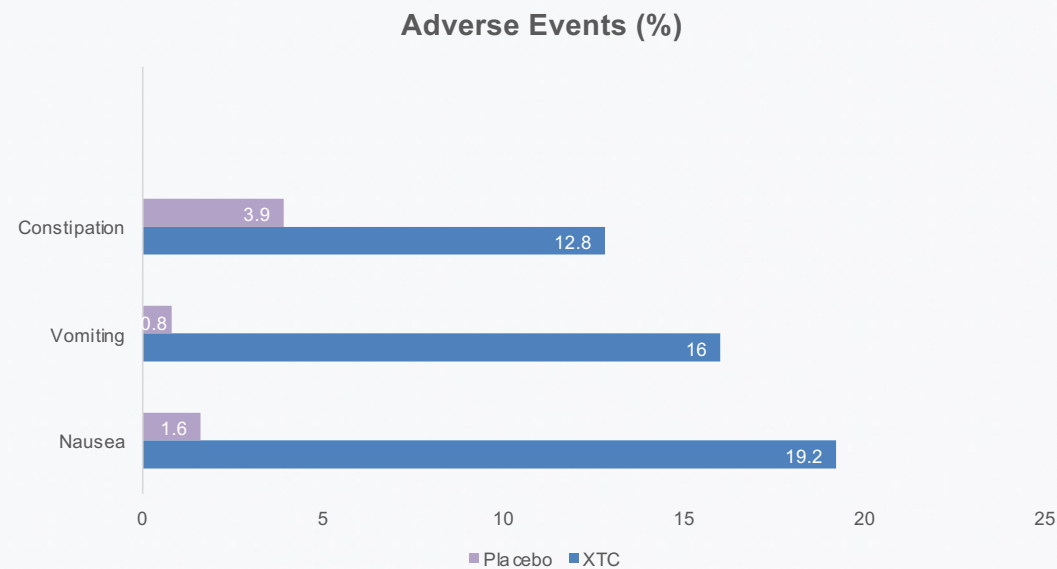
Effect size (Cohen's d) of **0.60** was observed at Week 5.

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Note the separation by Week 2. XTC achieved a significant PANSS reduction that was enhanced through the end of the trial (Week 5), showing robust clinical efficacy..

⁹Figure 2. Positive and Negative Syndrome Scale (PANSS) and Clinical Global Improvement-Severity (CGI-S) End Points (Modified Intent-To-Treat [mITT] Population) Reproduced from Kaul I, Sawchak S, Walling DP, et al. Efficacy and safety of xanomeline-trospium chloride in schizophrenia: a randomized clinical trial. *JAMA Psychiatry*. 2024;81(8):749-756. doi:10.1001/jamapsychiatry.2024.0785. Licensed under CC BY-NC-ND 4.0. Unaltered image.

Most Common Adverse Events



Majority of GI events were **mild-to-moderate** and **transient**, occurring in the titration phase

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Tolerability is key. While GI effects are higher than placebo, they rarely led to discontinuation. Most patients habituate to these effects within the first two weeks.

Differentiating the Safety Profile



Absence of Typical Side Effects

-  **Metabolic:** No significant weight gain compared to baseline.
-  **EPS:** No clinically significant akathisia or drug-induced parkinsonism.

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This is the differentiator for MSLs. We aren't just looking at efficacy; we're looking at the "removal" of the traditional antipsychotic burden: no weight gain, no movement disorders.

Clinical Response at Week 5

50.6%

Responder Rate (XTC)

Significant Patient Response

Defined as $\geq 30\%$ improvement in floor-adjusted PANSS scores.

Compared to only **25.3%** in the placebo group, XTC doubled the likelihood of a clinically meaningful response within 5 weeks.

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A 50% responder rate in an acute setting is highly impressive. It suggests that XTC is not just statistically better, it provides a 'noticeable' change in half the patients.

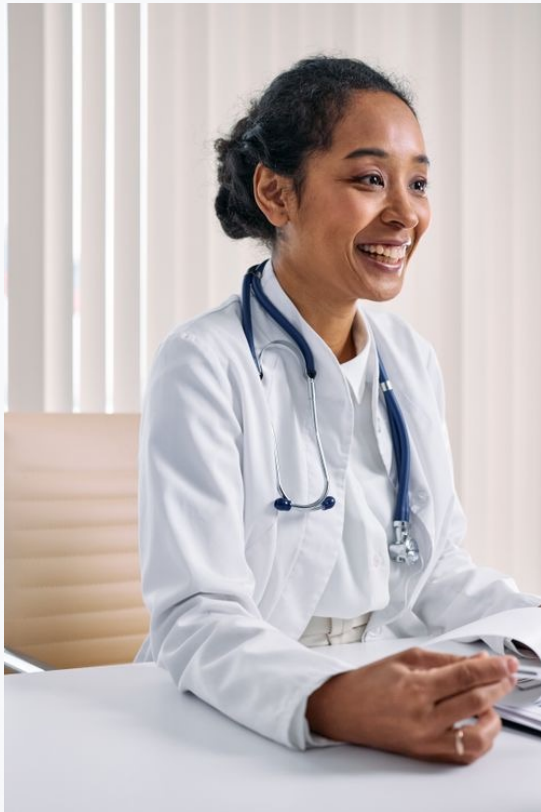
Conclusion & Future Outlook

- ★ **First-in-Class:** XTC is the first muscarinic agonist to show Phase 3 efficacy in schizophrenia.
- ★ **Broad Efficacy:** Addresses both positive and negative symptoms (PANSS subscales).
- ★ **Tolerability:** Avoids the metabolic and EPS burden of current antipsychotics.
- ★ **Strategic Value:** Represents a fundamental shift toward non-dopaminergic treatment pathways.

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In summary, XTC offers a new tool for clinicians to treat the whole patient – addressing the psychosis while protecting the patient's physical and metabolic health.

Questions & Discussion



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Thank you for your attention. I am happy to address any question regarding the EMERGENT program.

References

1. Huhn M, Nikolakopoulou A, Schneider-Thoma J, et al. Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of adults with multi-episode schizophrenia: a systematic review and network meta-analysis. *Lancet*. 2019;394(10202). doi:10.1016/s0140-6736(19)31135-3
2. Cerveri G, Gesi C, Mencacci C. Pharmacological treatment of negative symptoms in schizophrenia: update and proposal of a clinical algorithm. *Neuropsychiatr Dis Treat*. 2019;15:1525-1535. doi:10.2147/NDT.S201726
3. Feber L, Peter NL, Chiochia V, et al. Antipsychotic drugs and cognitive function: a systematic review and network meta-analysis. *JAMA Psychiatry*. 2025;82(1):47–56. doi:10.1001/jamapsychiatry.2024.2890
4. Chow RTS, Whiting D, Favril L, Ostinelli E, Cipriani A, Fazel S. An umbrella review of adverse effects associated with antipsychotic medications: the need for complementary study designs. *Neurosci Biobehav Rev*. 2023;155:105454. doi:10.1016/j.neubiorev.2023.105454
5. Wessel, Pham V, Vuckovic Z, et al. Xanomeline displays concomitant orthosteric and allosteric binding modes at the M4 mAChR. *Nat Commun*. 2023;14(1). doi:10.1038/s41467-023-41199-5
6. Guay DR. Trospium chloride: an update on a quaternary anticholinergic for treatment of urge urinary incontinence. *Ther Clin Risk Manag*. 2005;1(2):157-167. doi:10.2147/tcrm.1.2.157.62912
7. Rovner ES. Trospium chloride in the management of overactive bladder. *Drugs*. 2004;64(21):2433-2446. doi:10.2165/00003495-200464210-00005
8. Meyer JM, Correll CU. Increased metabolic potential, efficacy, and safety of emerging treatments in schizophrenia. *CNS Drugs*. 2023;37(7):545-570. doi:10.1007/s40263-023-01022-7
9. Kaul I, Sawchak S, Walling DP, et al. Efficacy and safety of xanomeline-trospium chloride in schizophrenia: a randomized clinical trial. *JAMA Psychiatry*. 2024;81(8):749-756. doi:10.1001/jamapsychiatry.2024.0785

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For those interested in the deep dive, I recommend starting with the Kaul 2024 primary paper which covers the full Phase 3 results in detail.